HMR INSTITUTE OF TECHNOLOGY & MANAGEMENT
Hamidpur, Delhi-110036
(An ISO 9001: 2008 certified, AICTE approved & GGSIP University affiliated institute)
E-mail:hmritmdirector@gmail.com, Phone: - 8130643674, 8130643690, 8287461931, 8287453693

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Weekly Progress Report (WPR)

WPR Dates: 16/1/2026 - 23/1/2026 **WPR of Week:**1

Name: Santosh Kumar

Enrollment Number: 12196502722

Class & Section: CSE-8A

Department: Computer Science and Engineering

Faculty Supervisor's Name: Ms. Indu Sharma

MAJOR PROJECT TITLE: - "AURA : Autonomous User Realization & Assembly"

TARGETS SET FOR THE WEEK:

- Define the problem statement and major objectives of AURA.
- Review related work on electronics inventory management, maker workflows, and component retrieval issues.
- Draft the high-level architecture for the host, node, inventory, and circuit workspace layers.
- Shortlist the initial hardware and software stack required for the prototype.
- Prepare the implementation direction and documentation roadmap for the coming weeks.

ACHIEVEMENTS FOR THE WEEK:

- Finalized the initial project scope around physical component localization and deterministic circuit assistance.
- Completed the early literature review and documented the major pain points in existing workflows.
- Prepared the first high-level architecture covering the host, distributed nodes, inventory layer, and workspace.
- Shortlisted the initial prototype stack using ESP32, nRF24L01+, WS2812 indication, and a structured software workflow.

- Created the first week-wise execution direction for implementation, documentation, and reporting.

FUTURE WORK PLANS:

- Begin designing the inventory data model and physical storage addressing strategy.
- Refine the host-to-node communication flow and physical location mapping logic.
- Draft the first UI flow for search, lookup, and component retrieval.
- Start planning the deterministic JSON model for circuit representation.

Week's Summary

Days/Time	5 Days a week Monday to Friday.
Monday	Defined the project problem, scope, and major objectives for AURA.
Tuesday	Reviewed related work on component management, maker workflows, and hardware retrieval challenges.
Wednesday	Sketched the first high-level architecture for host, node, inventory, and circuit workspace modules.
Thursday	Compared hardware options and shortlisted ESP32, nRF24L01+, WS2812, and low-power node control choices.
Friday	Organized reference notes, prepared the execution plan, and completed the first weekly progress documentation.

STATUS OF RESEARCH PAPER: Initial literature review started; foundational references on cyber-physical electronics assistance, component inventory workflows, and deterministic circuit representation were identified. Estimated overall project completion by the end of this week: 11%.

Supervisor signature: **Date:**